

Ratio and Proportion — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- To share a total: **add the parts, divide the total by that sum, then multiply back up.**
- For proportion questions, **find one first, then scale.**
- Give every ratio in its simplest form, and convert to a common unit before simplifying.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – UNITS, THE 1 : N FORM AND SCALE

1. Write each ratio in its simplest form.

(a) 250 g : 1 kg [1]

(b) 40 cm : 1.5 m [1]

(c) 20 minutes : 1 hour [1]

2. Write each ratio in the form 1 : n. The value of n may be a decimal.

(a) 4 : 10 [1]

(b) 5 : 12 [1]

3. A map is drawn to a scale of 1 : 50 000.

- (a)** Two towns are 6 cm apart on the map. Find the real distance between them, in kilometres. [2]

- (b)** A straight road is 8 km long. How long is it on the map, in centimetres? [1]

The multiplication is the easy mark. Converting centimetres to kilometres is the one that goes missing.

SECTION 2 – DIRECT PROPORTION

- 4.** 7 identical books have a total mass of 2.1 kg. Find the mass of 12 of these books. [2]

- 5.** A car uses 6 litres of fuel to travel 90 km. How far can it travel on 15 litres? [2]

- 6.** One pound sterling is worth ₹105.

- (a)** Convert ₹840 into pounds. [1]

- (b)** Convert £15 into rupees. [1]

7. Washing powder is sold in two sizes. [3]
Pack A: 900 g for ₹135. **Pack B:** 1.2 kg for ₹174.
Work out the price per 100 g of each pack, and state which is better value.

SECTION 3 – WORD PROBLEMS

8. Share 4500 in the ratio 2 : 3 : 4. [3]

9. A fruit drink is made by mixing orange juice and water in the ratio 2 : 7. A jug [2]
holds 1800 ml of the drink. How much orange juice does it contain?

10. In a car park the ratio of cars to motorbikes is 5 : 2. There are 24 more cars than [3]
motorbikes. How many motorbikes are in the car park?

You have not been given the total. Work out how many parts the 24 represents.

11. A recipe for 8 biscuits uses 120 g of butter and 180 g of flour.
(a) How much butter is needed for 20 biscuits? [2]

(b) Amit has 270 g of flour and plenty of butter. How many biscuits can he make? [1]

12. The ratio of Meera's age to Nikhil's age is 4 : 7. Nikhil is 21 years old. How old is Meera? [2]

End of paper. Check every answer has a unit or a form the question actually asked for.